SayedMorteza Malaekeh

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Education

University of Texas at Austin (UT Austin)
PhD Candidate (ABD) in Sustainable Systems Engineering — *Jan. 2023 – Exp. Dec. 2026*
Joint M.Sc. in Economics, conferred May 2026 (GPA: 3.9/4, 12 Ph.D.-level courses)

California Institute of Technology
Visiting PhD Researcher (Special Student) in Economics and Computer Science Groups — *May–Aug. 2024; May–Dec. 2025*

Sharif University of Technology
M.Sc. in Water Resources Engineering (Thesis A+; GPA: 4/4, Ranked First) — *Sep. 2019 – Feb 2022*

Sharif University of Technology
B.Sc. in Civil Engineering with Minor in Economics (GPA: 3.93/4) — *Sep. 2015 – Aug. 2019*

Selected Work/Research Experience

PhD Data Science Intern | Roblox Corporation — May – Aug. 2026

- Improved the Roblox Moments short-form video recommender system by redesigning ranking objectives to optimize long-term retention instead of short-term engagement.
- Applied double machine learning, causal forests, and deep learning models to petabyte-scale behavioral data, improving long-term-retention ranking by 30%+.
- Designed Bayesian optimization for A/B testing across short- and long-run outcomes to efficiently identify and validate improved recommender-system ranking objectives.
- Built production-ready ML and data pipelines over billions of impression and session events using Spark SQL/Hive and Trino, collaborating with ML, software, and data engineers.

Quantitative Research Fellow | Quanta Ventures Fund — Jan. – Mar. 2026

- Engineered daily trading strategies for a multi-strategy engine across major ETFs (QQQ, SPY, GLD) using ML and deep learning, reaching Sharpe and Calmar ratios of ∼2.0.
- Developed and backtested intraday models on minute-level ETF data to time entry and exit signals.
- Designed alpha signals combining mean reversion, momentum, and breakout logic, with ADX regime filters for adaptability across market environments.

Visiting PhD Researcher | California Institute of Technology — May-Aug. 2024; May-Dec. 2025

- Built a multimodal causal framework integrating deep learning and causal inference.
- Integrated Vision Transformers, CNNs, and autoencoders with R-Learner and Causal Forest.
- Developed a new estimator for conditional average treatment effects with image-based treatments, applied to large multimodal datasets in environmental policy.

Graduate Researcher | Lawrence Berkeley National Laboratory — June 2024 – June 2025

- Designed a nationwide propensity-score matching strategy balancing solar PV adopters and non-adopters.
- Led acquisition of a $180K Experian credit dataset covering 10M households, 2010–2023.
- Engineered scalable pipelines to clean and analyze 13+ TB of monthly household-level credit data.
- Applied synthetic control and staggered DiD to estimate financial impacts of solar PV adoption.

Fellowships, & Awards

2026 **Academy Distinguished Alumni Graduate Fellowship**, UT Austin
2025 **Robert Abbasi & Shahnaz Hemmati Graduate Fellowship**, UT Austin
2024 **Grad Camp Fellowship**, UC Berkeley & **Travel Grant**, Princeton University
2020 **Exchange Program Scholarship**, Sharif Univ. of Technology
2019 **Ranked 9th/10,000 (top 0.1%)** — Iranian National Economics Olympiad

Selected Publications (Google Scholar)

- **4 peer-reviewed journal articles + 1 peer-reviewed NeurIPS Workshop paper**; 59 citations; published in *Energy Policy*, *Agricultural Economics*, *Environmental Research Communications*, and *SERRA*; additional manuscript under review at *Nature Energy.*

Skills

Programming: Python (PyTorch, Scikit-learn, XGBoost), SQL, R, MATLAB, STATA
ML/AI: Recommender Systems, Deep Learning, LLMs (Hugging Face, LangChain, PEFT/LoRA), Bayesian Optimization
Causal/Experimentation: A/B Testing, Causal Machine Learning, DiD, DML/R-Learner, Causal Forests, Synthetic Control, Regression Discontinuity, EconML
Data/Systems: Spark/PySpark, Hive, Trino, Databricks, AWS, GCP, Docker, Kubernetes, HPC